

Retrieval Augmented Generation

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Agenda

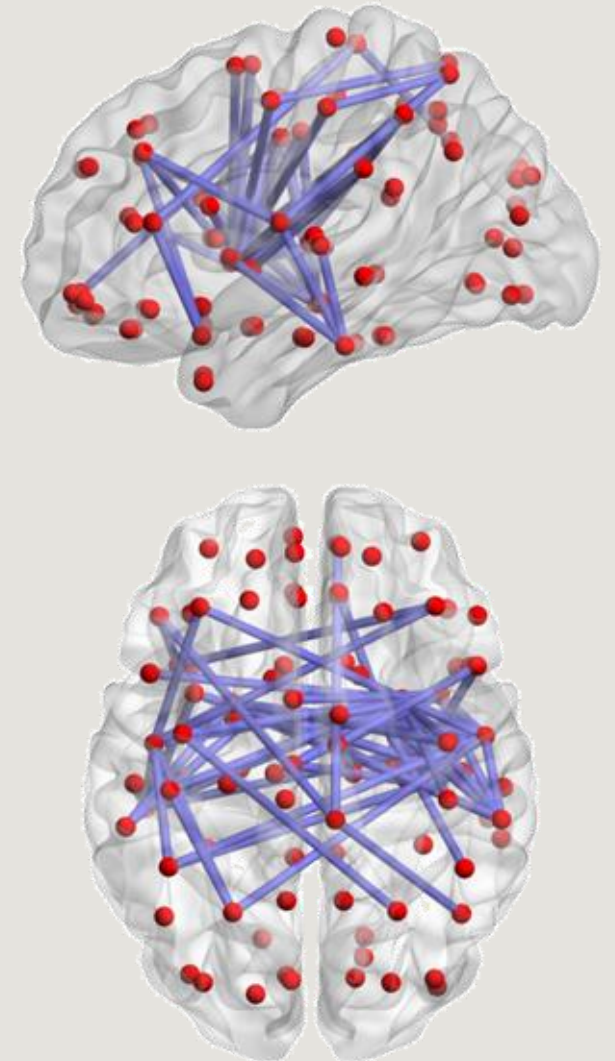
Recap of Neural Networks

Intuition behind GenAI

Intro to Natural Language Processing

Large Language Models

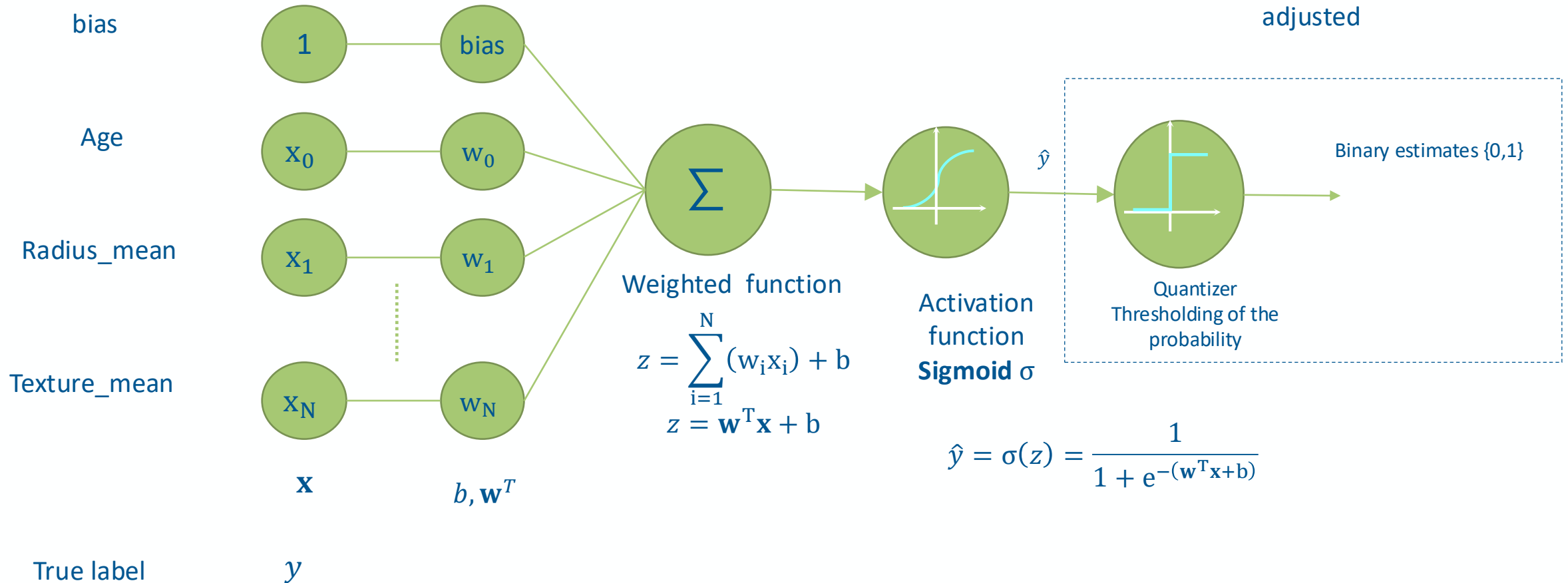
Retrieval Augmented Generation



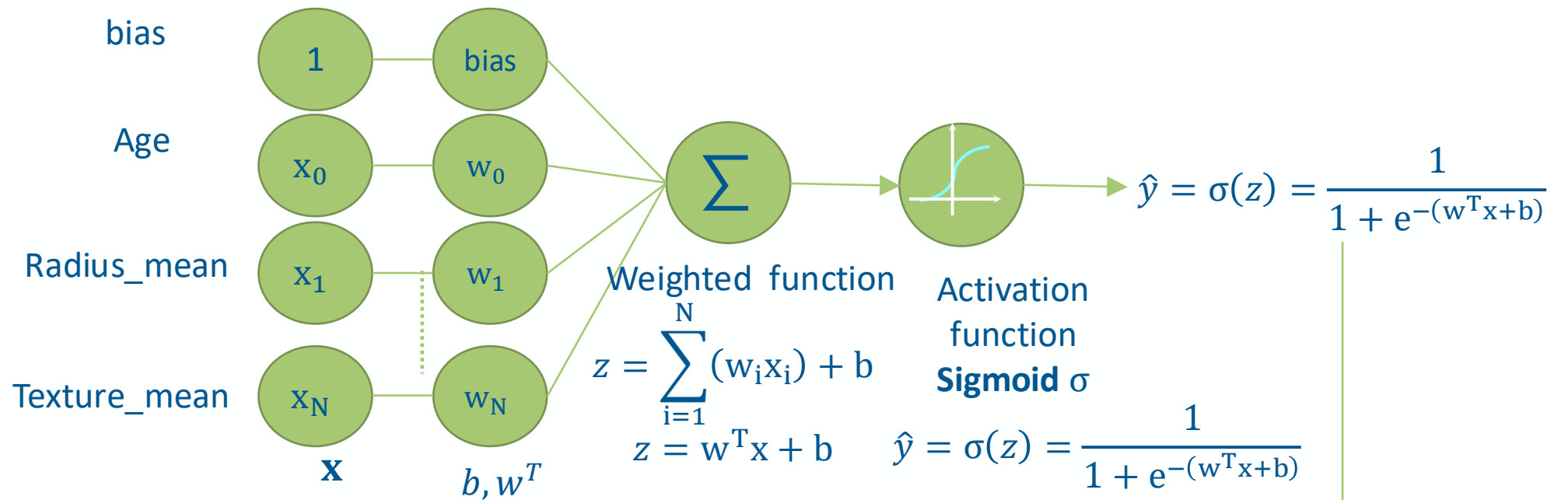
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From regression to logistic regression

We do this with specific threshold to convert probabilities to predictions, the threshold can be adjusted



Logistic Regression to predict breast cancer using features extracted from pathological images



Benign 0
malignant 1

$y^{<i>}$

$$J(\omega, b) = -\frac{1}{M} \sum_{i=1}^M (y^{<i>} \log(\hat{y}^{<i>}) + (1 - y^{<i>}) \log(1 - \hat{y}^{<i>}))$$

Gradients of the cost (error) with respect to the parameters (weights and bias)

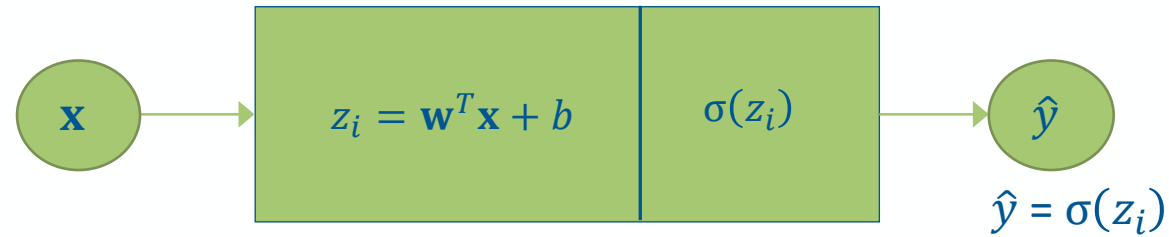
$$\omega : \omega - \alpha \frac{\partial}{\partial b} J(\omega, b), \quad \omega : \omega - \alpha \frac{\partial}{\partial b} J(\omega, b),$$

Probability of Benign
probability of malignant

where:

- $y^{<i>}$ is the true label (0 or 1) for the ($i - th$) data point.
- $\hat{y}^{<i>}$ is the predicted probability from our model for the ($i - th$) data point

Gradient Descent for Logistic Regression



Cost function:

$$J(\boldsymbol{\omega}, b) = -\frac{1}{M} \sum_{i=1}^M (y^{<i>} \log(\hat{y}^{<i>}) + (1 - y^{<i>}) \log(1 - \hat{y}^{<i>}))$$

To find the optimal weights, we need the gradient of the cost function with respect to (w) . To do so, we apply the chain rule:

$$\frac{\partial J}{\partial \boldsymbol{\omega}} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial \boldsymbol{\omega}}$$

SoftMax for multinomial classification

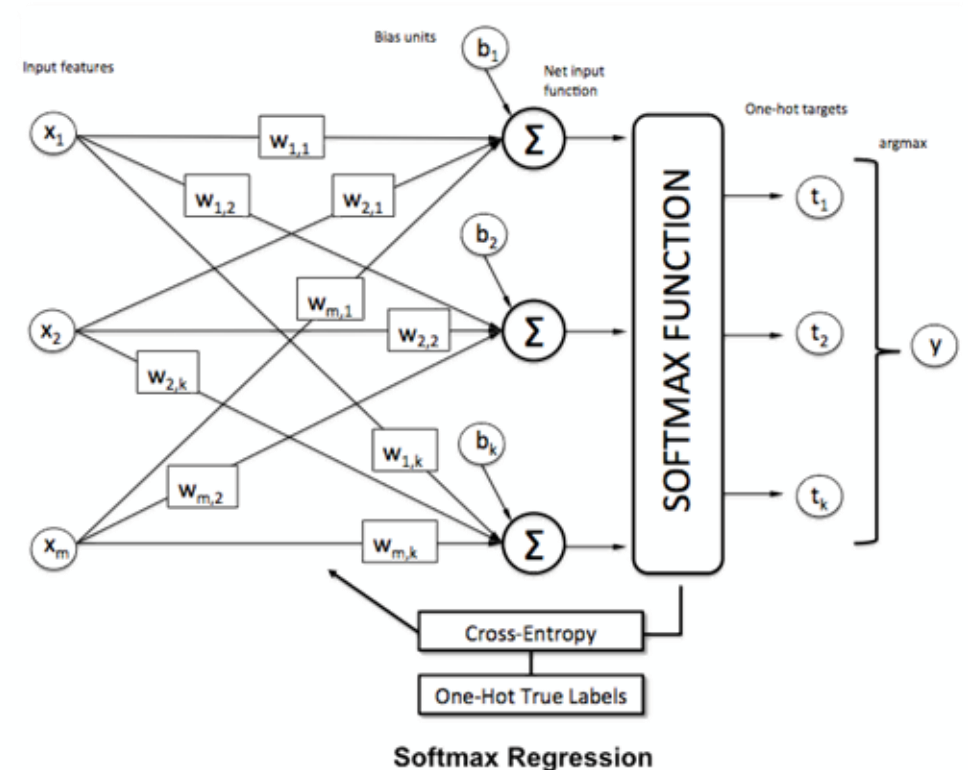
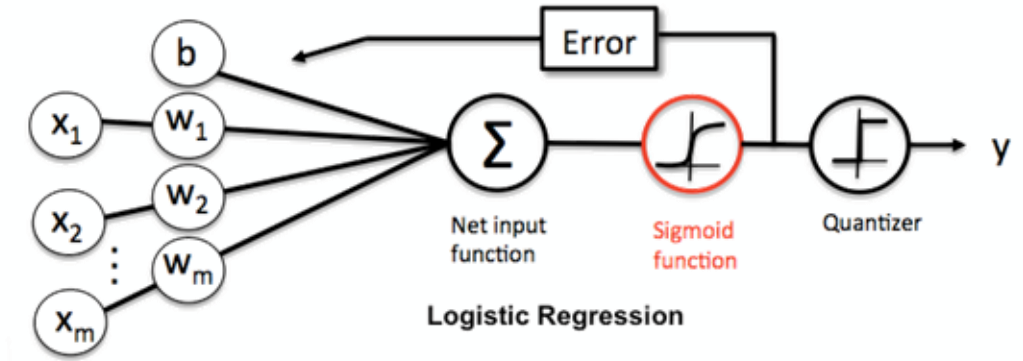
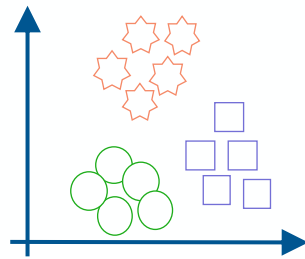


We use the SoftMax function and fit a SoftMax regression model

→ **SoftMax**

Normalized exponential function (probabilities of different classes), maps a vector of continuous values to a vector of probabilities summing to 1

$$\begin{bmatrix} 1.30 \\ 5.10 \\ 2.20 \\ 0.70 \\ 1.10 \end{bmatrix} \rightarrow \text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \rightarrow \begin{bmatrix} 0.02 \\ 0.90 \\ 0.05 \\ 0.01 \\ 0.02 \end{bmatrix}$$

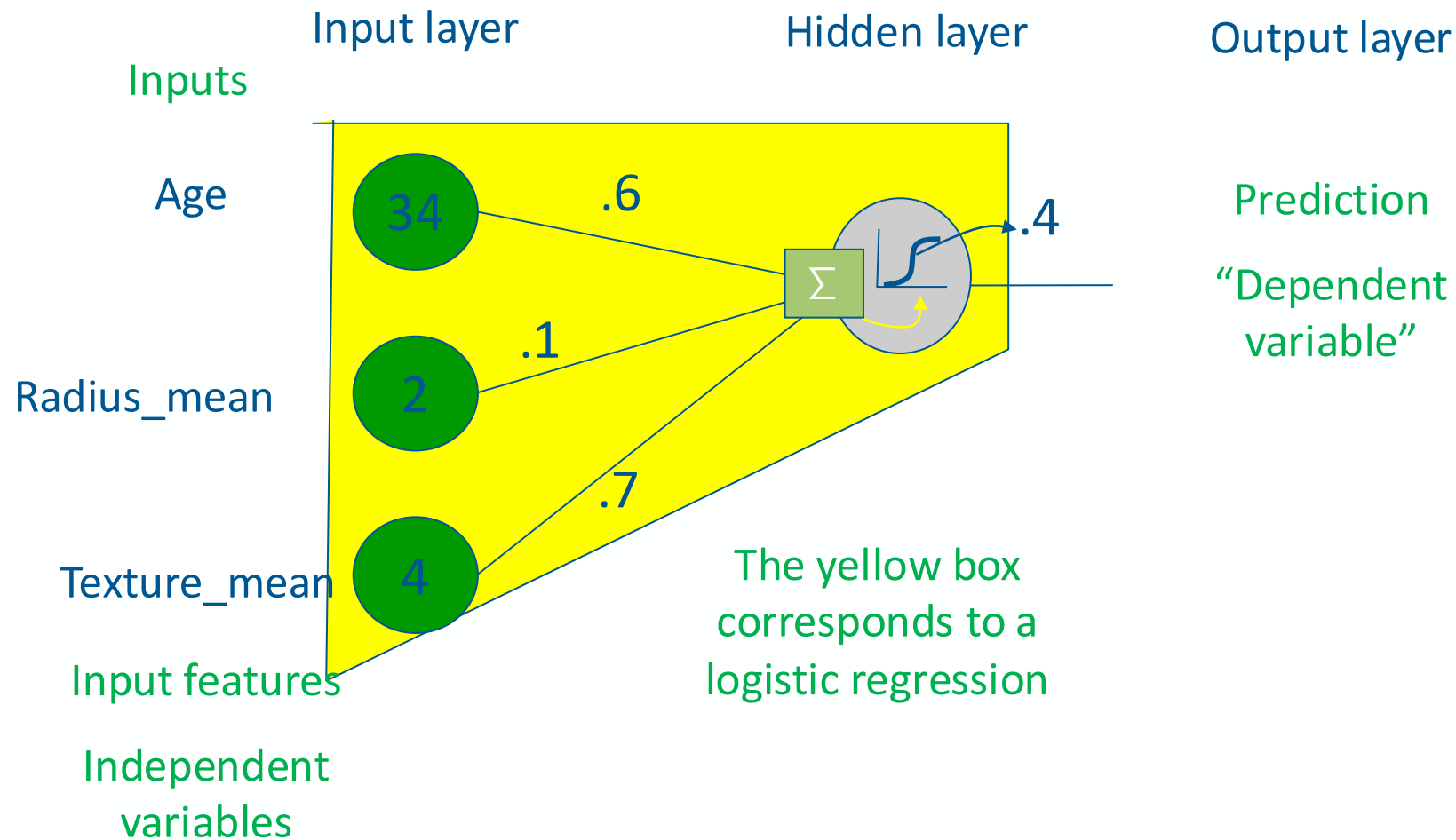


We still don't have a good result using the Logistic regression model

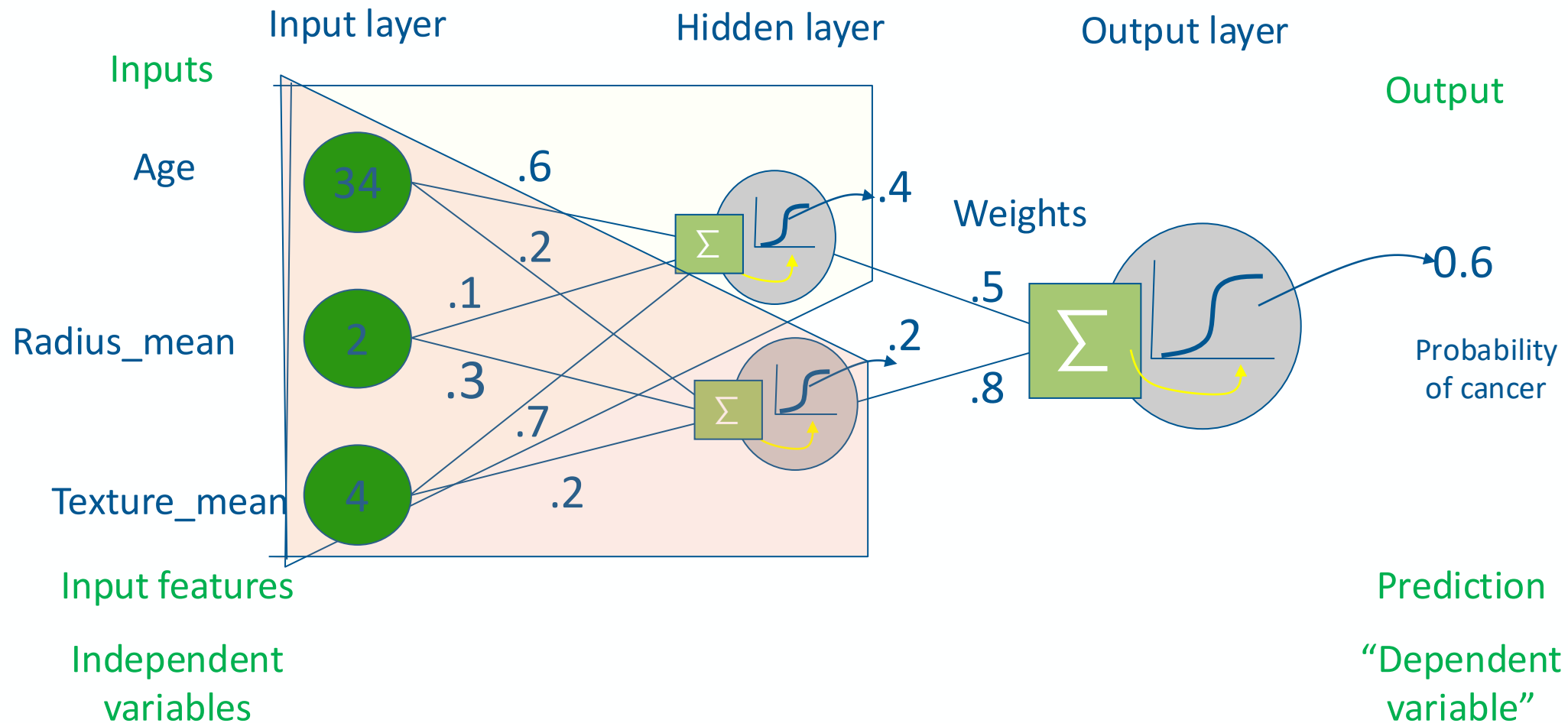
The relationship between input and output is a complex one and cannot be modelled with such a simple model?



What about adding a hidden layer to model a complex relationships between input and output

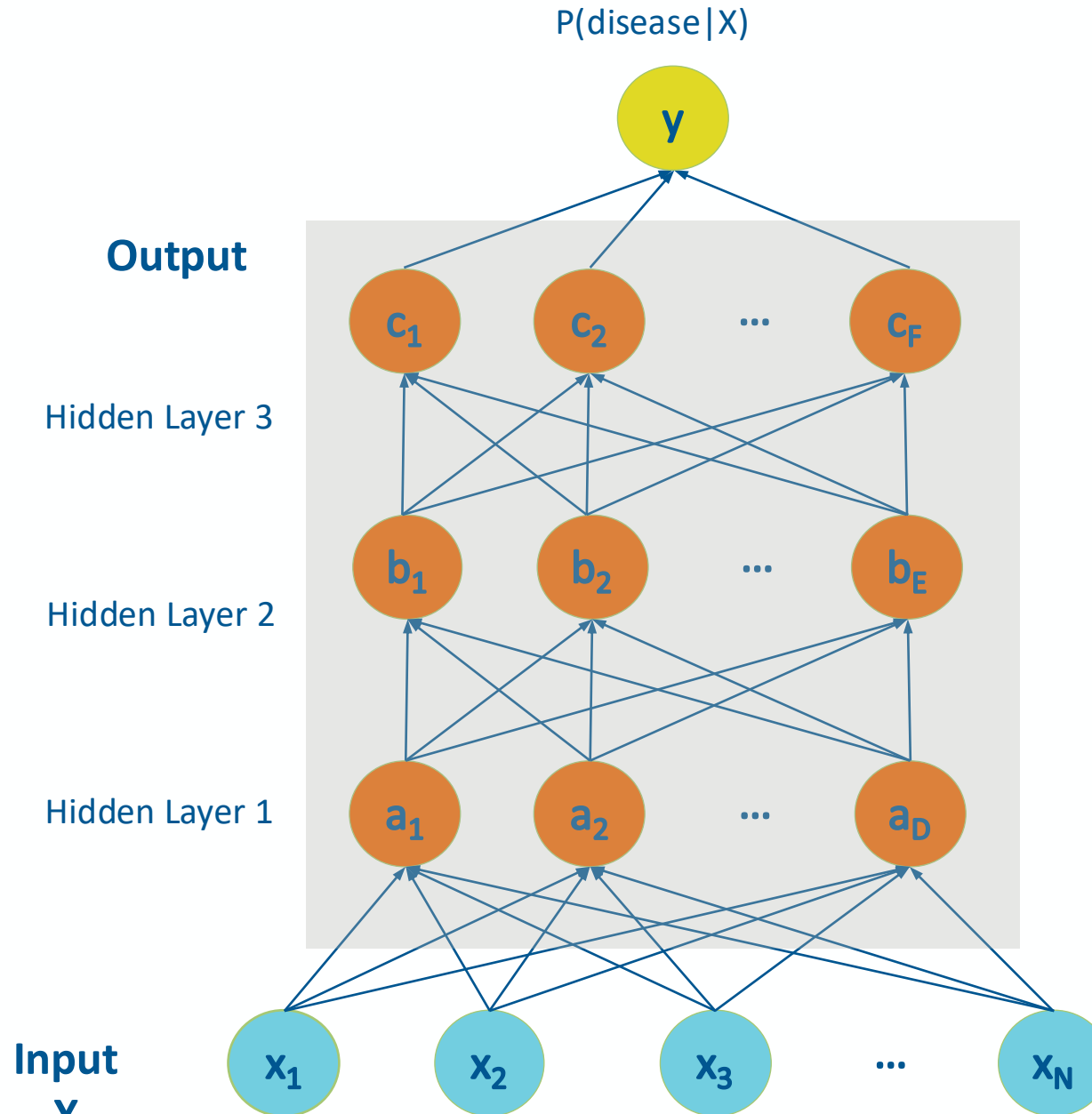


Adding a hidden layer to model a complex relationships between input and output



Bottomline: neural network can be explained as several logistic regression blocks—with some caveats and extensions. We will call this blocks now neurons in a neural network.

Binary classification (e.g. illness vs. Healthy of specific disease)

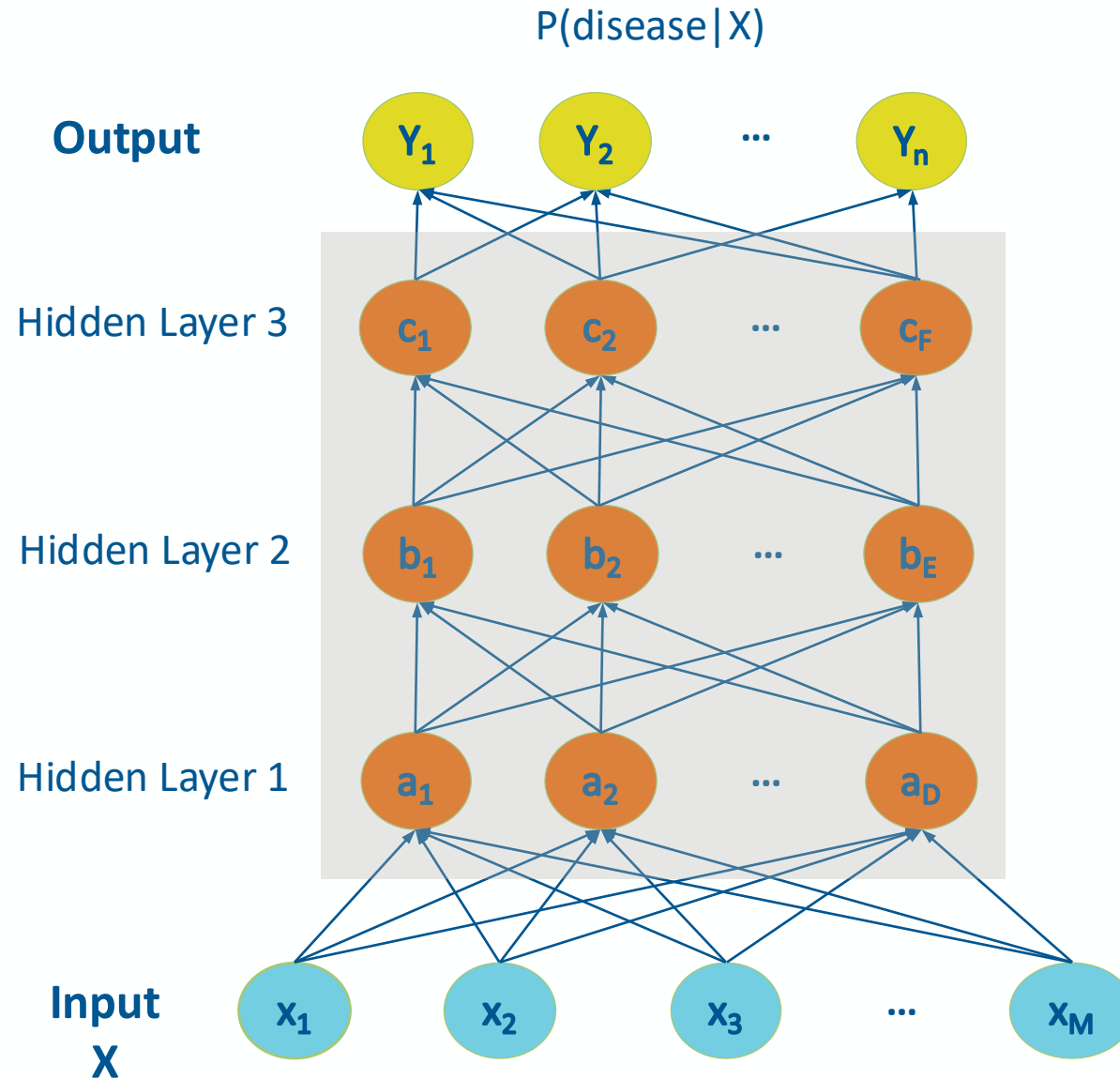


To train a neural network, we need:

- training data
- an objective function to assess performance and guide learning

Before we start

Disease prediction: Multinomial classification



Generative AI: the intuition behind!

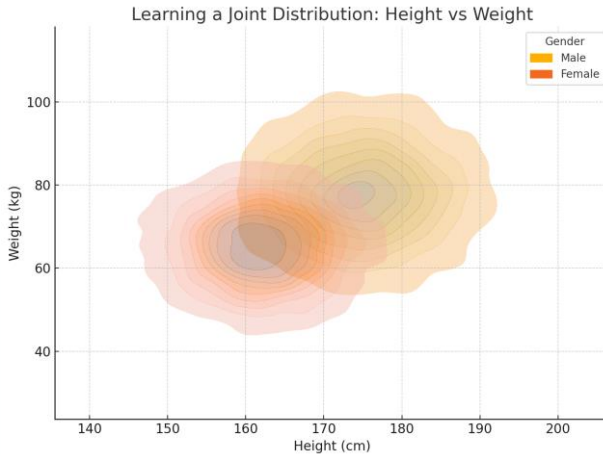


AI-generated image

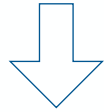
What do we need to generate realistic images with Neural Networks?

- To generate something we need an architecture that can generate (e.g. we need output layer that can output images)
- We need some kind of stochasticity. We don't want to get the same output each time we run the model.
- We need a proper objective function to use in the optimization and to find the parameters
- We need a NN will **learn the patterns and regularities** — i.e., it approximates the **underlying data distribution**.
- We should make sure that the generated data adheres to the same training data distribution
- The generated data point could be then considered as samples from the approximated data distribution

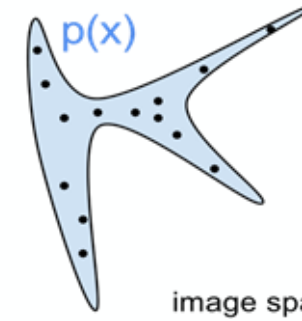
In GenAI, Generalization Requires Accurate Distribution Learning



sampling

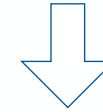


- Male 190, and 90 kg is realistic sample
- Female 140 and 20 kg is not a realistic sample



High dimensional distribution of $3 \times 1024 \times 1024$ pixels' values of students ID pictures: $P(X)$

sampling



This is not a realistic sample This is not a realistic sample

Large Language Models



Before 2020, who attempted to utilize Language Models?

Language model in Action

[Web](#) [Images](#) [Videos](#) [Maps](#) [News](#) [Shopping](#) [Gmail](#) [more](#) ▼



 **Everything**

▼ **More**

Any time

Past 24 hours

▼ **More search tools**

i love you because you |

i love you because you **love me**

i love you because you **are**

~~i love you because you are you~~

i love you because you gave me sausage and cheese

~~i love you because you understand me lyrics~~

i love you because you **understand me**

i love you because you **are you poem**

i love you because you **look like jim reeves**

i love you because you **are the perfect definition baby**

i love you because you **are the perfect definition**

Text is a Sequence data

Sentiment
classification

"The series I've enjoyed most in my
life thus far."



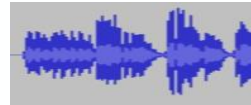
Machine
translation

"Artificial intelligence is useful"



"La inteligencia
artificial es útil"

Speech
recognition



Transcript

Speech synthesis

Written
text



DNA sequence
analysis

"ATGCTTCGGCAAGACTCAAAAATA"



Properties

Music generation

∅



Unlike other sequence data formats, text is more difficult to represent

- **Unstructured:** Text lacks a predefined format or consistent numerical structure, unlike tables or time series.
- **Ambiguous:** Words can have multiple meanings depending on usage (e.g., "bank" as a financial institution or riverbank).
- **Context-dependent:** The meaning of text relies on surrounding words and broader context (e.g., "cold" could refer to temperature or illness).

Different tasks in Natural Language Processing NLP

Sentiment Analysis

- **Goal:** Determine the emotional tone or opinion in text.
- **Input:** “smells amazing, good purchase!” **Output:** Positive, negative, or neutral sentiment.

Summarization

- **Goal:** Condense a longer text into a shorter version while retaining key information.

Question Answering (QA)

- **Goal:** Answer questions based on a given context or knowledge base.
- **Example:** “When was Pepsi founded?”

Named Entity Recognition (NER)

- **Goal:** Identify and classify proper nouns (e.g., people, organizations, places).
- **Example:** “Apple Inc. was founded by Steve Jobs.” → Apple Inc. (ORG), Steve Jobs (PERSON)

Text Classification

- **Goal:** Assign categories or labels to text.
- **Example:** Spam detection, topic classification.

Machine Translation

- **Goal:** Translate text from one language to another.
- **Example:** English to French: “Hello” → “Bonjour”

Text Generation

- **Goal:** Produce new, coherent text based on input prompts.
- **Used in:** Chatbots, storytelling, creative writing.

All these NLP Tasks Can Be Framed as “Text In → Text Out”

A general-purpose learning strategy for language can provide a foundation (skeleton) for most of these applications

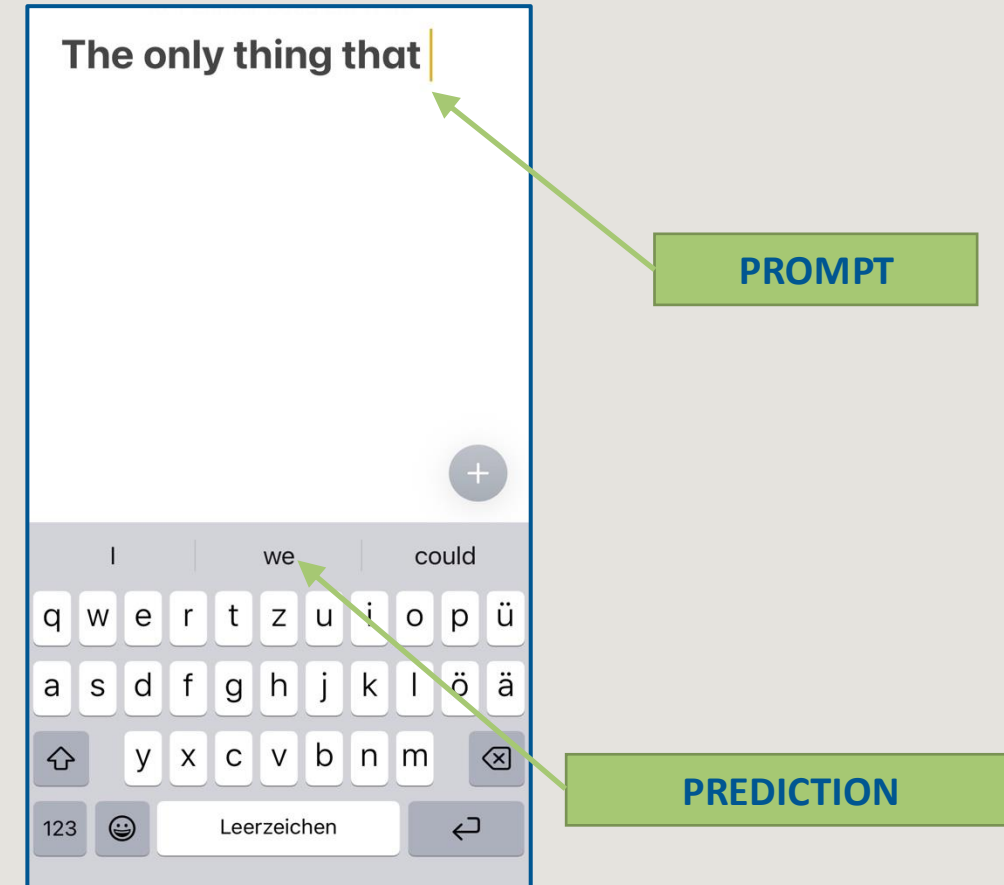
Language modelling is the general-purpose learning strategy for language

- A Language model is a model trained to **understand human language** by predicting the next word in sequence:

$$P(x_t \mid x_1, x_2, \dots, x_{t-1})$$

- **Self-Supervised training**
 - Training an LLM requires **no manual labels** — it learns from raw text by treating parts of the text as input and the next word as the target.
 - This makes it **scalable** to massive datasets (e.g., web data, books, code).
 - This process teaches the model Syntax (grammar, word order) and Semantics (meaning)

A language model in action



Context to predict next token

Context is the text before the question

Honey, can you
pass me the _____



What does he
mean?

Conditional Probability of the Missing Word

$P(x|\text{context}) = P(\text{salt} \mid \text{Salad again, really! and she forgot to season, Honey, can you pass me the })$



Context to predict next token

Honey, can you
pass me the ____

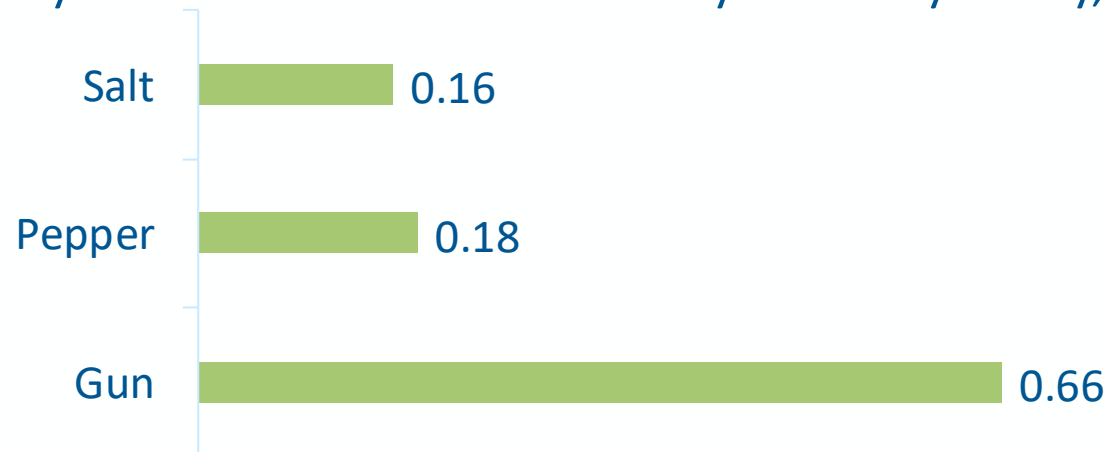
Context is the text before the question



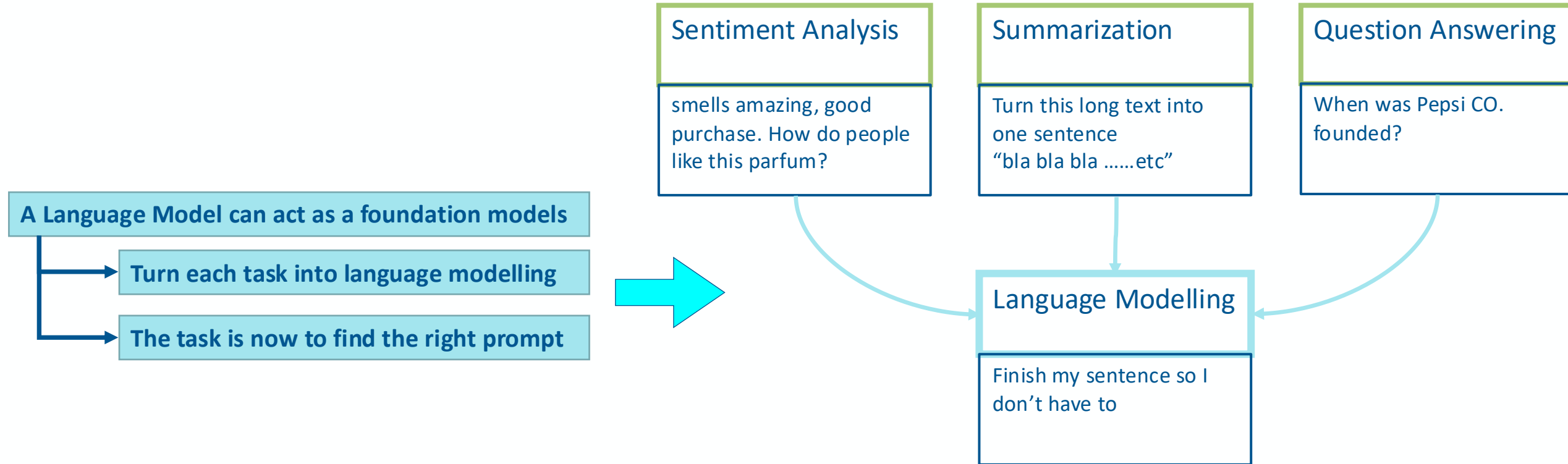
What does he
mean?

Conditional Probability of the Missing Word

$P(x|\text{context}) = P(\text{salt} \mid \text{nobody ever robs restaurants. I love you honey bunny, Honey, can you pass me the })$



What makes language models so important?



How to Train a Large Language Model (LLM): Simplified Steps

Step 1: Collect a Large Text Corpus

- Gather massive amounts of text data from diverse sources:
 - **Common Crawl:** A broad web crawl dataset containing billions of web pages.
 - **Wikipedia:** English and possibly other language versions.
 - **Books:** Public domain books (e.g., from Project Gutenberg) and possibly licensed ones.
 - **WebText:** A dataset created by OpenAI (for GPT-2) from web pages linked from Reddit with a high karma score.

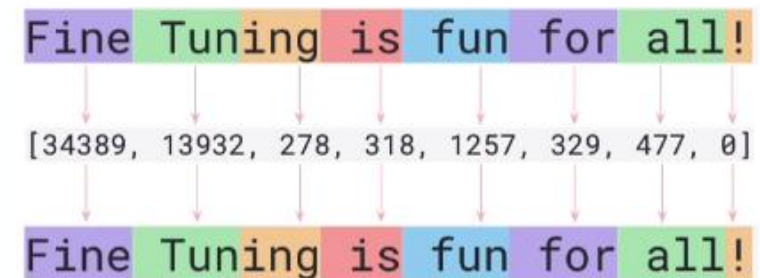
How to Train a Large Language Model (LLM): Simplified Steps

Step 2: Tokenization

- LLMs don't read raw text — they work with numbers, and tokenization is the first step toward converting text into numbers. Therefore, we need to break all the text into small chunks called **tokens**.

- These can be:

- ✓ Words: "Hello", "world"
- ✓ Subwords: "play", "##ing"
- ✓ Characters or bytes (depending on the tokenizer)

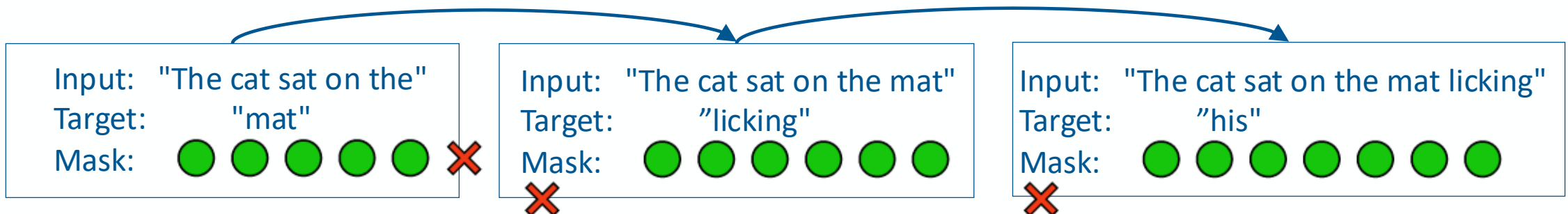


- The output is tokens (numbers) that represent each word in the vocabulary of input data
- The larger the vocabulary size the more unique tokens we will have in our lookup table**

How to Train a Large Language Model (LLM): Simplified Steps

Step 3: Predict the Next Token (Training Loop)

- Train the model to **predict the next token** in a sequence:
 - Take a sentence, and randomly hide part of it (truncate or mask)
 - The model guesses the missing token (missing word) based on the rest of the sentence
 - The model uses **attention** architecture to understand which words relate to which.
 - Compare the guess to the actual (ground truth) token
 - Adjust the model's weights to improve future predictions (via backpropagation)
- Over time, the model learns grammar, facts, style, and reasoning — all by learning to complete sentences well.



Causal Masking: Learns to generate next token and is used to train LLM in self-supervised manner

How to Train a Large Language Model (LLM): Simplified Steps

Step 4: Repeat Over the Whole Corpus

- This process is repeated **billions of times** across the entire dataset, until the model can generalize patterns in human language.

This process called **Self-supervised learning (SSL)** is a machine learning approach where a model learns from data without explicit labels.

Disclaimer: Training an LLM is **computationally expensive** — it requires thousands of GPUs and weeks of training time. That's why companies like OpenAI, Google, and Meta train these models and fine-tune them for specific tasks later.

Quiz



- What does an LLM output on its output layer?

- A softmax function can be used at the output layer of an LLM since it provides probabilities of different tokens

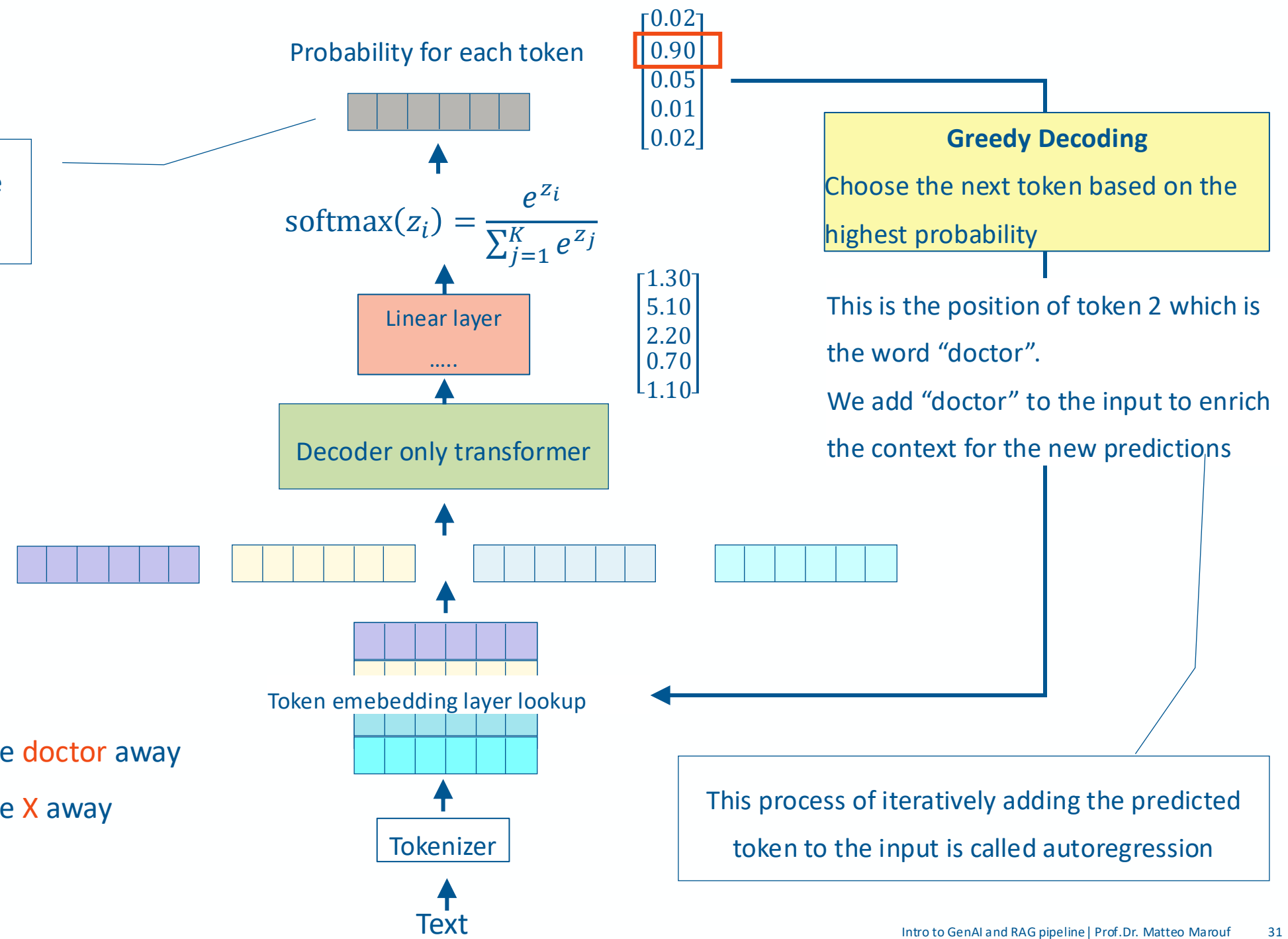
Quiz



- What is the size of the output layer in the LLM?

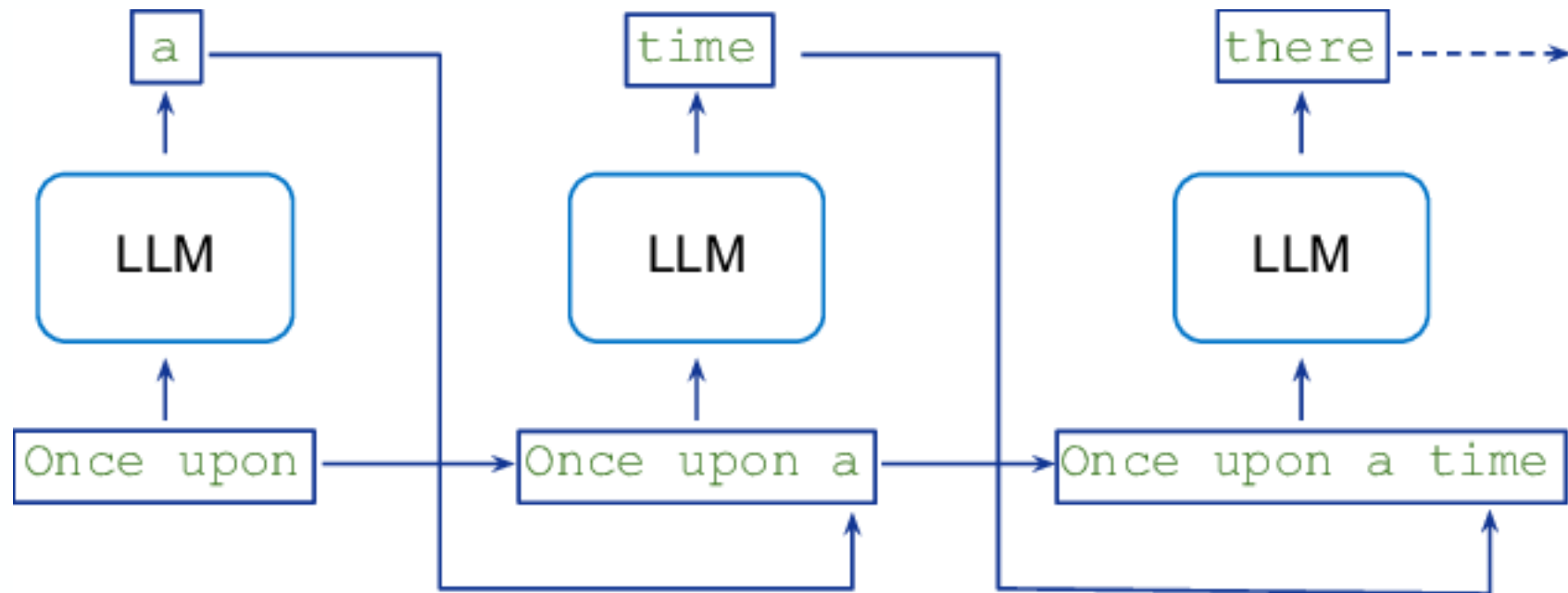
- Vocabulary size or total number of tokens

one apple a day keeps the **doctor** away
 one apple a day keeps the **X** away



Autoregression is then used to generate new text

Each time the model generates/predicts a token, that token is added to the prompt to create a new context



Remember:

- The model will receive tokens and not words at its input
- The model will then predict a vector of probabilities that correspond to different tokens
- The size of the output layer is related to the size of the vocabulary.

Quiz

- Given this architecture, will the model predict the same tokens for two identical inputs?
- Yes
- No



How to ensure stochasticity and creativity of the answers if we always predict the most probable next token ?

If you always pick the **most probable next token** (called **greedy decoding**), the model will:

- Produce **safe**, repetitive, and **deterministic** responses.
- Lack **variety or imagination** in output.
- Often **loop or repeat phrases**.



We need a stochastic and a controllable sampling

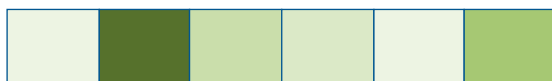
Temperature-based Sampling

The **temperature** controls how “sharp” or “flat” the probability distribution is:

- $T = 1.0$: default distribution
- $T < 1.0$: sharper, more deterministic (less creative)
- $T > 1.0$: flatter, more randomness (more creative)

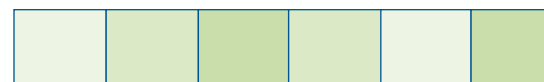
$$p(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Softmax with $T=1$



$$p(z_i) = \frac{e^{\frac{z_i}{T}}}{\sum_j e^{\frac{z_j}{T}}}$$

Softmax with T (e.g. 2)



- The temperature-controlled sampling controls diversity of the answers, creativity, stochasticity
- Once the probabilities are adapted, we can apply sampling. For example, sample the top k



Chatbots like ChatGPT is a product that uses LLM to interact with users.

* AI-generated image

ChatGPT is an application built using a large language model (LLM).

To turn an **LLM** into a **chatbot like ChatGPT**, several key steps are added:

- **Prompt Engineering:** Designing system messages and prompts to guide behavior (e.g., tone, style, role).
- **Fine-Tuning / Alignment:** Training the model further with human feedback (RLHF) to follow instructions and stay safe.
- **Memory & Context Handling:** Managing conversation history and summarizing context to maintain coherence.
- **UI/UX Integration:** Creating a user-friendly interface for chatting (web app, mobile app, API).
- **Tool Integration:** Connecting to tools like web search, code interpreters, or image generation.
- **Safety & Moderation Layers:** Filtering harmful content and enforcing usage policies.

Retrieval-augmented Generation

You know what they call a Quarter Pounder with Cheese in Paris?



Pitfalls of LLM-based chatbots



Static Knowledge

LLMs are trained on fixed datasets and cannot access new or changing information after their training date



Hallucination

LLMs may confidently generate **inaccurate or fabricated answers**, especially on niche topics.



Context Window Limits

LLMs have a maximum token limit (e.g., 128K tokens for GPT-4o), restricting how much content can be passed in.



Domain Knowledge Gaps

Generic pretraining means LLMs may lack depth in **company-specific or technical domains**.



No Source Attribution

LLMs don't show where their answers come from.



Low Explainability

Hard to verify if an answer is accurate or how it was generated.

Retrieval Augmented Generation (RAG) vs Fine Tuning

RAG and **Fine Tuning** are two pivotal approaches to customizing Large LLMs for domain-specific tasks.

RAG

leverages external information sources at query time, dynamically retrieving the most relevant documents before passing them to the LLM for response generation.

It doesn't require altering the underlying model, which makes it resource-efficient and easy to update.

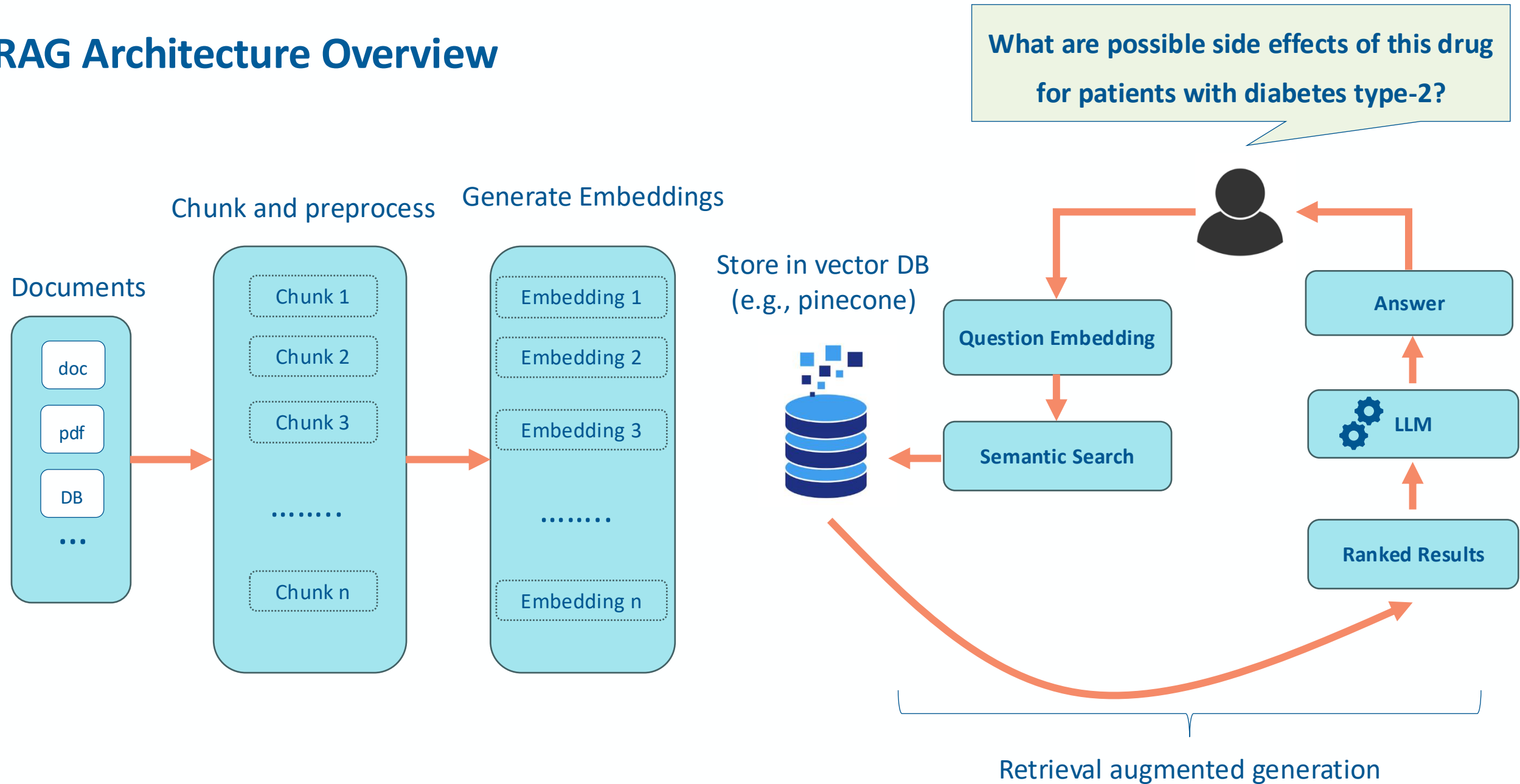
Fine Tuning

involves modifying the internal parameters of a pre-trained model by training it on a domain-specific dataset.

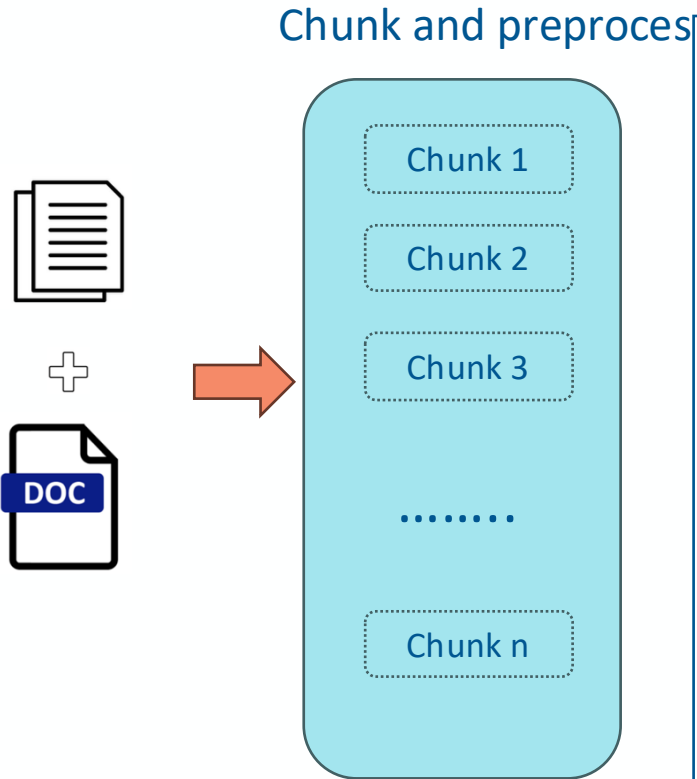
This method produces a tuned model capable of generating highly tailored outputs without querying an external database.

RAG offers agility and adaptability—ideal for evolving datasets—while fine tuning provides deep optimization for stable domains where precision and consistency are paramount.

RAG Architecture Overview



Chunking and preprocessing



Purpose: Split large documents into smaller pieces for better retrieval.

Challenges: Preserve context, avoid chunking mid-sentence or across semantic units.

Tools:

- **LangChain:** RecursiveCharacterTextSplitter, MarkdownHeaderSplitter
- **LlamaIndex:** custom node parsers
- **Haystack:** preprocessing pipeline

Each chunk will contain the relevant text with necessary metadata to improve context awareness

Chunk:

Employees working more than 30 hours per week are eligible for full health insurance coverage. Coverage includes medical, dental, and vision, with premiums partially paid by the company. Enrollment must be completed within 30 days of hire or qualifying life event.

Metadata:

```
{  
  "document_title": "Employee Benefits Guide 2024",  
  "source": "HR Portal",  
  "page_number": 5,  
  "section": "Health Insurance",  
  "date_uploaded": "2024-01-10",  
  "tags": ["benefits", "insurance", "hr"],  
  "doc_id": "benefits_2024_001"  
}
```

```
1 text_splitter = RecursiveCharacterTextSplitter(  
2     separators = [",", " ", " "], # need to put separator based on text  
3     chunk_size = 50,  
4     chunk_overlap = 20,  
5     length_function = len,  
6     add_start_index = True  
7  
8 )  
9 #Split the long unstructured string  
10 chunks = text_splitter.create_documents(colorado_raw)  
11 chunks[:5]
```

Splitting text by recursively look at characters from LangChain

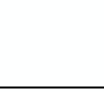


Purpose: Turn text chunks into vectors for semantic search.

Tools (Embedding Models):

- **OpenAI:** text-embedding-3-small, text-embedding-3-large
- **Cohere, Anthropic, Google** (Universal Sentence Encoder)
- **Open-source:** BGE, Instructor-XL, E5, MiniLM

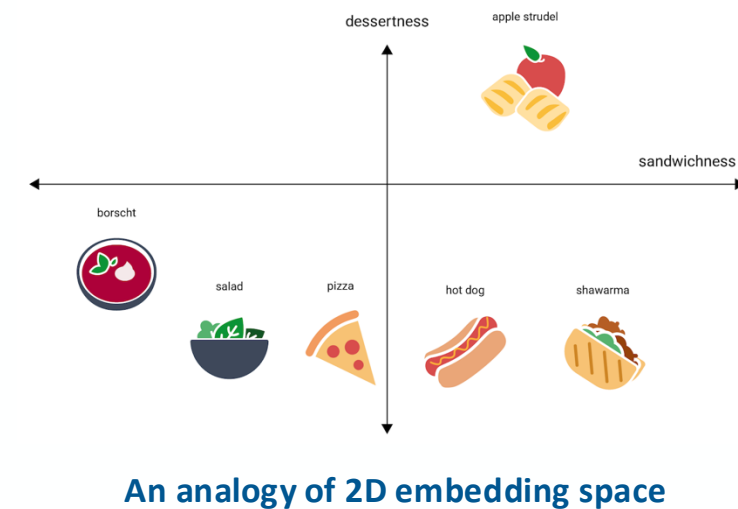
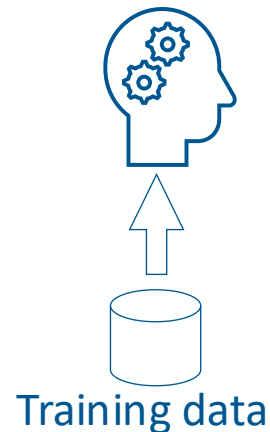
Intuition: Words Embedding

borscht
[1, 0, 0, 0, ..., 0]
     
hot dog
[0, 1, 0, 0, ..., 0]
     
shawarma
[0, 0, 0, 0, ..., 1]
     

Source: <https://developers.google.com/machine-learning/crash-course/embeddings>

One-hot encodings of borscht, hot dog, and shawarma

An embedding approach will use the training data to find useful representation that can be used to evaluate and study the semantics of words/tokens



- Meanings or relationships are captured and more explicit in the embedding space
- **RAG uses Sentence embeddings** which are essentially an extension of word embeddings, aiming to encode the meaning of a sequence of words rather than just individual words.

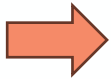
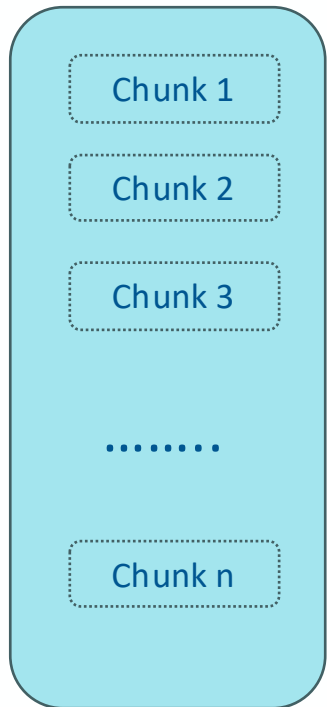
Embedding Models

Model	Size	Training Data	Speed (Approx.)	Embedding Dimension	STS Benchmark	Best Performance
all-MiniLM-L6-v2	22 MB	1 billion pairs	~5,000 sentences/s	384	~76.65%	Lightweight, efficient for bias detection and semantic tasks
all-mpnet-base-v2	110 MB	1 billion pairs	~2,500 sentences/s	768	~78.90%	High accuracy for bias classification tasks
all-roberta-large-v1	350 MB	2.5 TB multilingual	~1,000 sentences/s	1024	~76%	Good for larger, multilingual datasets but slower
xlm-roberta-base	550 MB	100+ languages	~800 sentences/s	768	~72%	Suitable for multilingual text, but less accurate for English

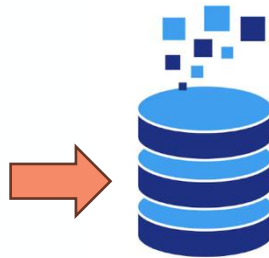
Vector Store (Indexing + Retrieval)

Chunk and preprocess

Generate Embeddings



Store in vector DB
(e.g., pinecone)



Purpose: Store and retrieve text embeddings efficiently using similarity search.

Tools:

- **FAISS** (Facebook AI Similarity Search) – local, fast, popular
- **Pinecone** – managed, scalable
- **Weaviate** – semantic graph & hybrid search
- **Qdrant** – open-source, high-perf
- **Milvus, Chroma** – open-source options

A Chunk in Pinecone

Vector values of chunk text



Update Vector

Namespace

(Default)

ID

42007ed6-fcf1-4db3-a47c-c5300f3cea6a

Values

0.00566215673,-0.0208694711,-0.0191473328,0.0592168942,0.038249306

+ Sparse data

Additional Metadata of each chunk



category

marketing

Aa

doc_type

PDF

Aa

file_name

Marketing_strategy_2020.pdf

Aa

page_num

1

123

text

- They rely heavily on - Weekly reports and campaign retrospectives are crucial for their continuous improvement.

Aa

+ Add metadata

Retrieval

What are possible side effects of this drug for patients with diabetes type-2?



Embeddings of Questions

Semantic Search to find the most relevant chunks



Vector DB

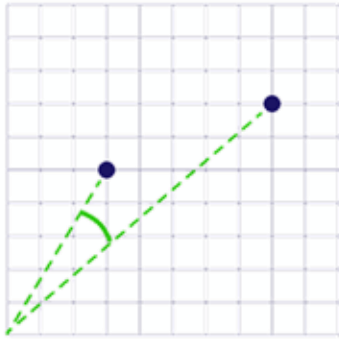
Some questions for thought:

- How do we find the most relevant (or semantically similar) chunks?



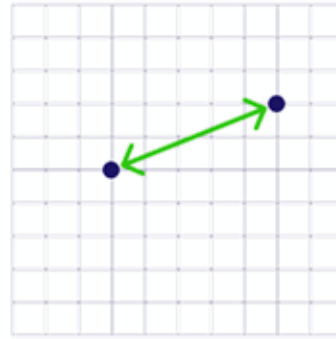
Retrieved Chunks from DB

In embedding space, we need distance Metrics as we work with vectors



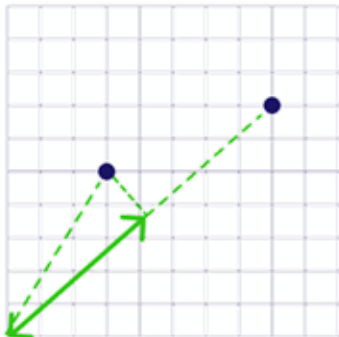
Cosine Distance

$$1 - \frac{A \cdot B}{||A|| \ ||B||}$$



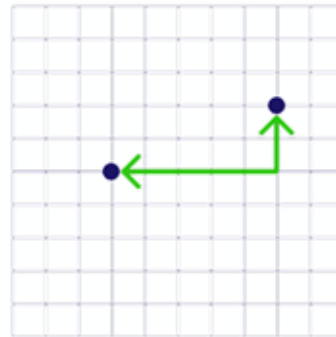
Squared Euclidean (L2 Squared)

$$\sum_{i=1}^n (x_i - y_i)^2$$



Dot Product

$$A \cdot B = \sum_{i=1}^n A_i B_i$$



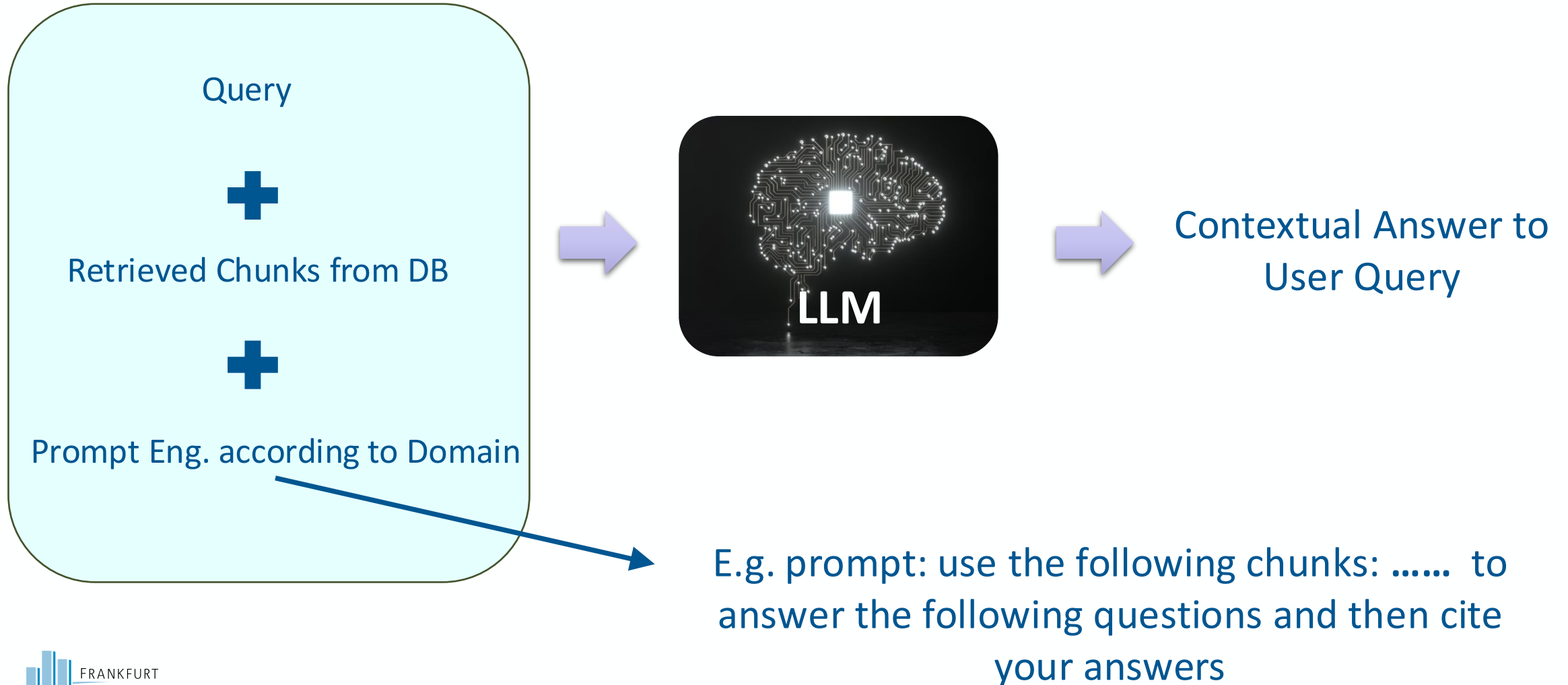
Manhattan (L1)

$$\sum_{i=1}^n |x_i - y_i|$$

- The **cosine distance** measures the angle between two vectors in a multi-dimensional space – with the idea that similar vectors point in a similar direction.
- The **dot product** shows the alignment of two vectors
- The **Euclidean distance or L2 norm** reflects the distance between each of the values of the vectors being compared.
- **Manhattan or the L1 norm** is calculated by taking the sum of the absolute values of the vector

The Generation in Retrieval Augmented Generation

we provide the retrieved chunks along with the question to an LLM to provide Contextual Answer



Example prompt that is retrieved, engineered, and sent to an LLM

[User Question]

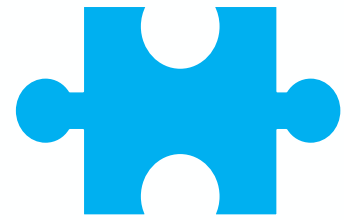
How does the Marketing team make strategic decisions?

[Context]

- The Marketing team recently shifted focus to data-driven decision-making.
 - They rely heavily on - Weekly reports and campaign retrospectives are crucial for their continuous improvement.
- A/B testing, customer segmentation, and performance metrics.

[Instruction]

Using the provided context, answer the question in a concise and informative way.



Questions



Answers

